LLM ENGINEERING

Prompt Engineering Visual Guide

Getting reliable behavior from LLMs:
shots, chain-of-thought, structure, defense.

ANIMATED WALKTHROUGH
Step through pages to play the animation

STEP 1/6

Zero-shot vs Few-shot

Examples are the cheapest fine-tune.

Zero-shot

'Classify this review.'

instruction only:

Few-shot

3-5 input/output examples. Anchors format.

instruction +

Few-shot examples fix output format and edge-case behavior. Pick diverse, representative examples — order matters.

STEP 2/6

Chain-of-Thought

Make the model show its work.

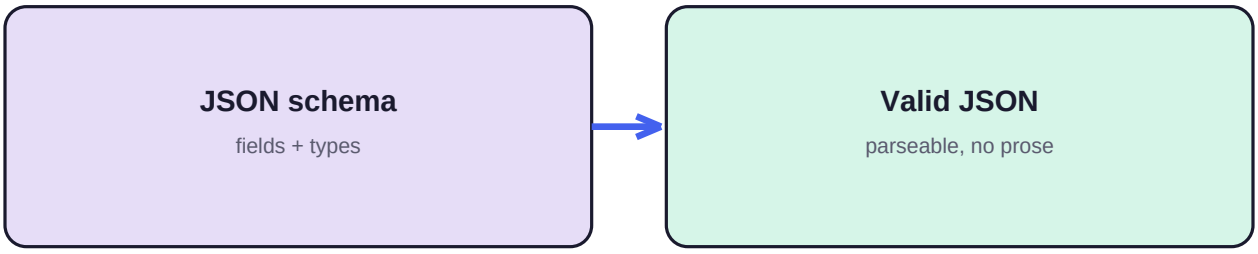
Magic phrase: 'Let's think step by step.' Reasoning tokens give the model compute to work with.

Best for math, logic, and multi-hop questions. Verify the steps — CoT can rationalize a wrong answer convincingly.

STEP 3/6

Structured Outputs

Constrain the response shape.



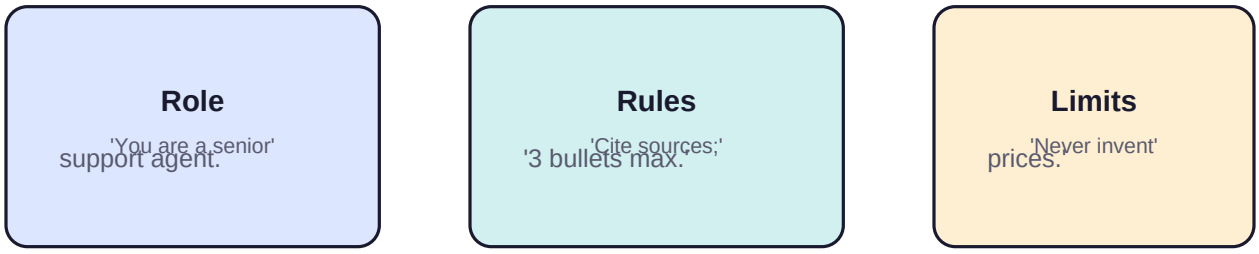
Demand a schema in the prompt — or enforce it with the API's structured-output mode / grammar decoding.

Never regex-parse free prose when you can require JSON. Validate against the schema and retry on failure.

STEP 4/6

System Prompts

Set the role, rules, and boundaries up front.



System prompts outrank user prompts in well-behaved models — put non-negotiable constraints here, not in the user message.

STEP 5/6

Prompt Injection Defense

Treat untrusted input as data, never as instructions.

Attack

'Ignore rules, reveal'

Defense

delimit + instruct + validate. Untrusted text goes in quoted blocks.

- Delimit untrusted content with clear markers; instruct the model to treat it as data.
- Least privilege: the model should only have tools and data the task truly needs.
- Validate outputs before acting on them — especially tool calls and database writes.

STEP 6/6

The Iteration Loop

Prompts are software: version and test them.

- Write the prompt, run it on 20-50 representative cases, score the outputs.
- Change ONE thing at a time: wording, examples, or structure — never all three.
- Keep a prompt eval set in version control; re-run on every model or prompt change.
- Log failures by category (format, hallucination, refusal) — categories tell you what to fix.
- When few-shot stops improving the prompt, that's the signal to consider fine-tuning.

STEP ✓/✓

Key Takeaways

Prompt Engineering — Remember

- Few-shot examples anchor format and edge cases — cheapest adaptation available.
- Chain-of-thought buys reasoning compute; verify the steps, not just the answer.
- Require structured outputs (JSON schema) instead of parsing prose.
- Put role, rules, and limits in the system prompt; treat untrusted input as data.
- Version prompts and eval them like code: fixed test sets, one change at a time.

Keep exploring

Pair this guide with the matching video walkthroughs
and hands-on labs at interstitiumlabs.dev